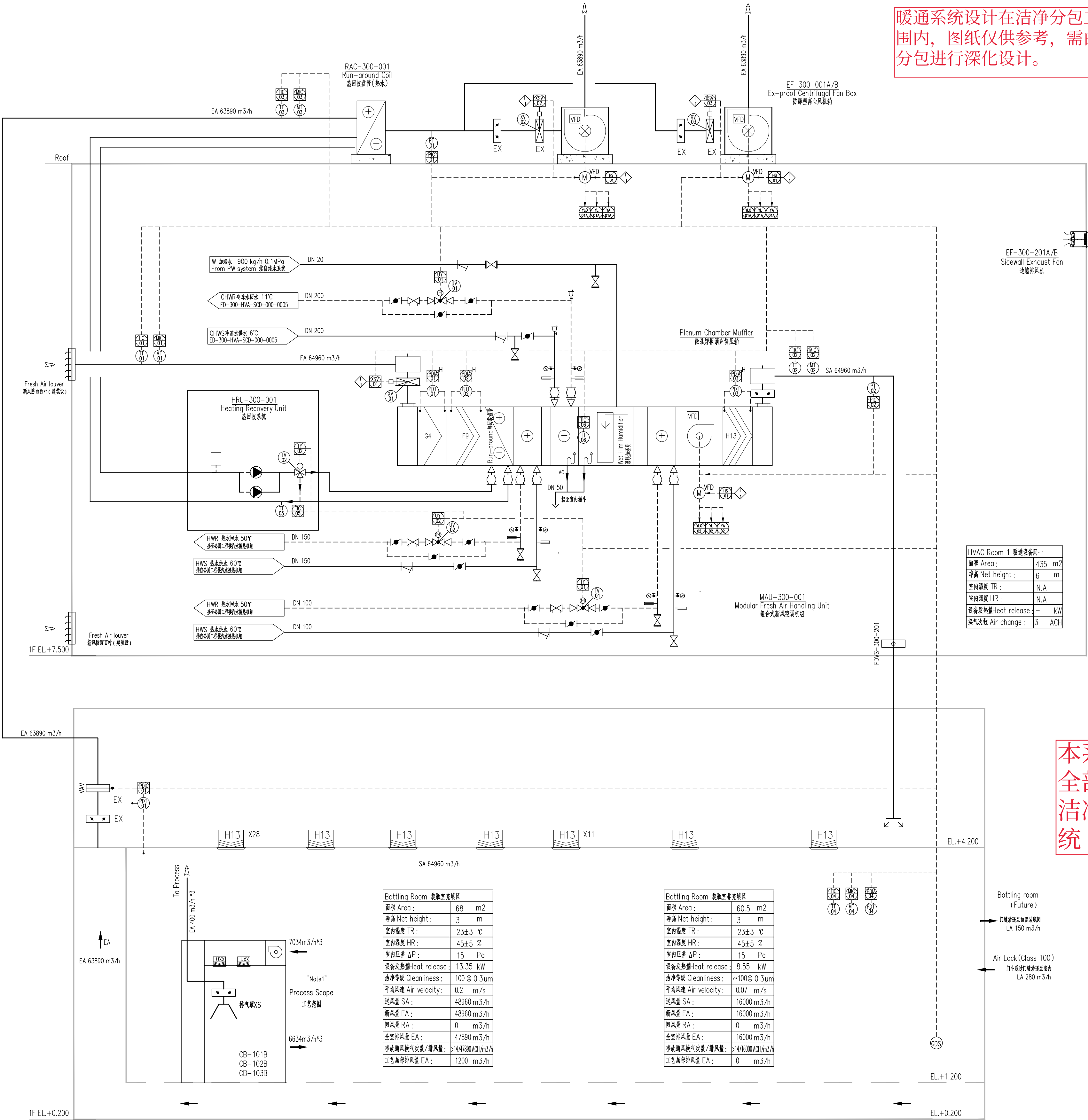


Abbreviation 缩写	Legend 图例	Description 说明
AD		Square Diffuser 方形散流器
SG		Single Deflection Grille 单层百叶风口
RK		Openable Single Return Grille With Filter 可开启回风口带滤网
L		Rain proof Louver 防雨百叶风口
WL		Exterior Walls Rainproof Louver 外墙防雨百叶
HEPA/ULPA		HEPA/ULPA 高效/超高效过滤器
FFU		Fan Filter Unite 风机过滤器单元
F/D		With Filter/Damper 风口带滤网/调节阀
		Multi-blade Damper 手动对开多叶调节阀
		Check Damper 止回阀
MD		Motorized On-off Damper 电动开关调节阀
FDVS		Normally Open 70°C Fire Damper 70°C常开防火调节阀
VAV		Variable Air Volume Damper 变风量阀
CAV		Constant Air Volume Damper 定风量阀
CAV		Excess Pressure Damper 余压阀
DCC		Reheating Coil with Condensate Water Pan 再热器盘管(带冷凝水收集盘)
RHC		Recooling Coil 再热器盘管(热水)
RAC		Run-around Coil 热回收盘管
		General Valve 通用阀门
		Globe Valve 截止阀
		Check Valve 止回阀
		Butterfly Valve 蝶阀
		Ball Valve 球阀
		Gate Valve 闸阀
		Electric Globe Valve/Electric Gate Valve 电动截止阀/电动闸阀
		Electric Butterfly Valve 电动蝶阀
		Differential Pressure Bypass Electric Valve 压差旁通电动调节阀
		Y Type Filter Y型过滤器
		Pressure Gauge 压力表
		Thermometer 温度计
		Pressure Reducing Valve 减压阀
		Balance Valve 平衡阀
		Safety Valve 安全阀
		Steam Trap 疏水阀
		Rubber Hose 橡胶软管
		Flexible Metal Hose 金属软管
		Automatic Exhaust Valve 自动排气阀
		Reducing Pipe 变径
		Temperature Sensor 温度传感器
		Humidity Sensor 湿度传感器
		Differential Pressure Sensor 压差传感器
		Duct Static Pressure Sensor 风管静压传感器
		Interlock 联锁
		Gas Detector 可燃气体探头
		Controlled by AC controller 受控空调控制系统



暖通系统设计在洁净分包工作范围内，图纸仅供参考，需由洁净分包进行深化设计。

本系统全部为洁净系统

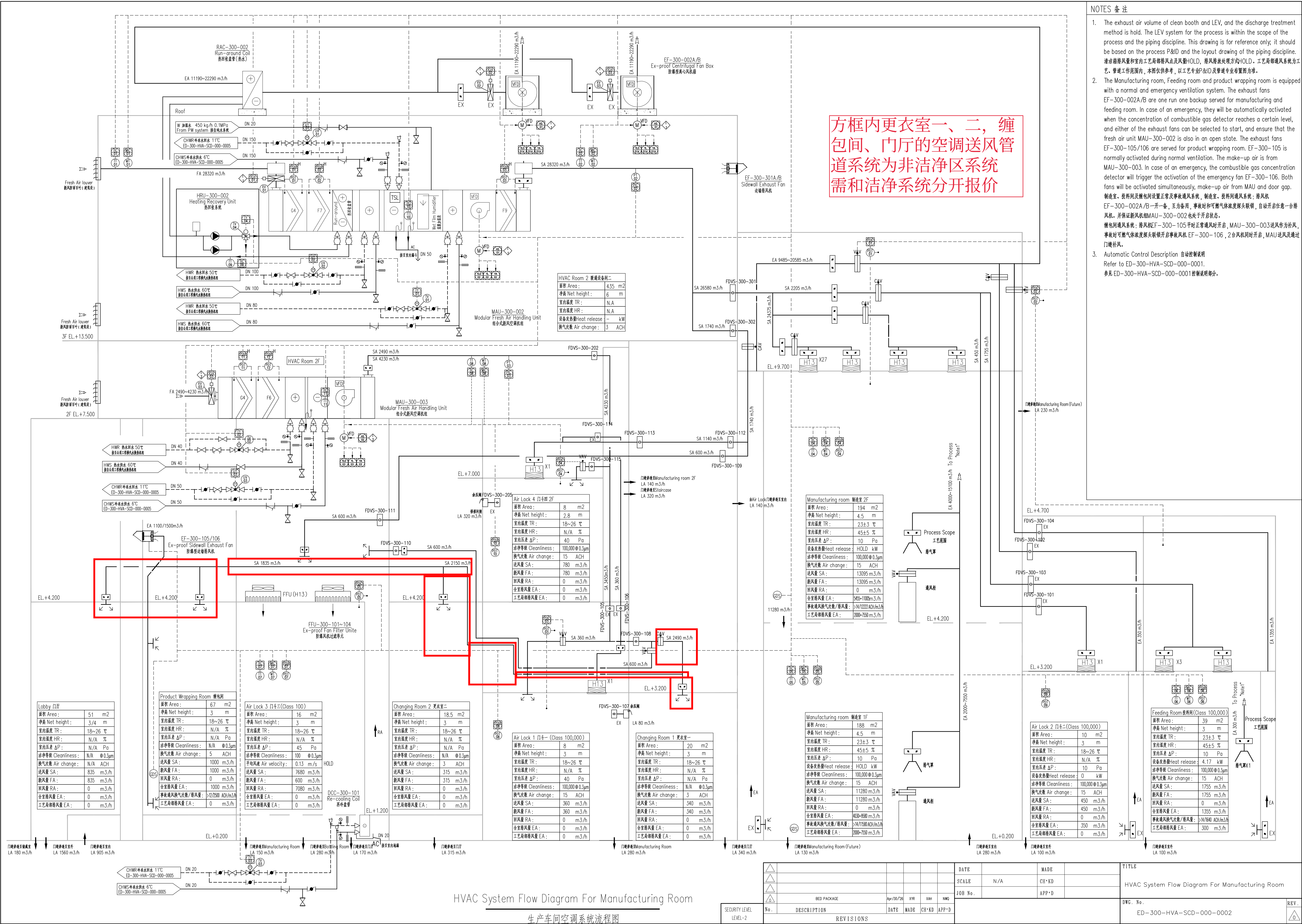
NOTES 备注

1. The LEV system for the process is within the scope of the process and the piping discipline. This drawing is for reference only; it should be based on the process P&ID and the layout drawing of the piping discipline. The air volume and simultaneous usage coefficient will be further detailed in the DD stage. 工艺局部通风系统为工艺、管道工作范围内，本图仅供参考，以工艺专业P&ID及管道专业布置图为准。具体风量及同时使用系数详设计阶段细化。

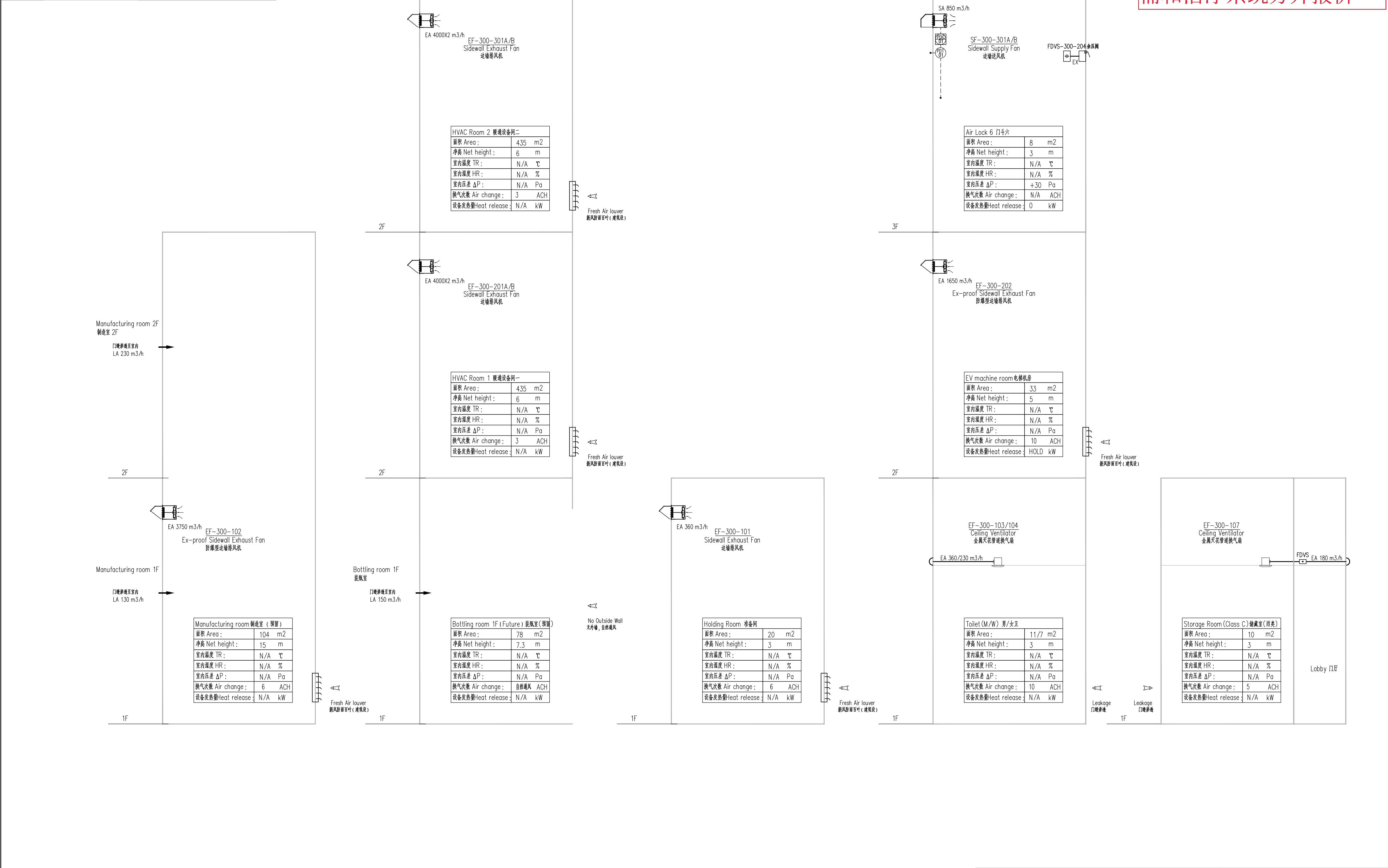
2. The bottling room is equipped with a normal and emergency ventilation system. The exhaust fans EF-300-001A/B are one run one backup. In case of an emergency, they will be automatically activated when the concentration of combustible gas detector reaches a certain level, and either of the exhaust fans can be selected to start, and ensure that the fresh air unit MAU-300-001 is also in an open state. 装瓶室设置正常及事故通风系统，排风机EF-300-001A/B一开一备，互为备用，事故时和可燃气体浓度探头联锁，自动开启任意一台排风机。并保证新风机组MAU-300-001也处于开启状态。

3. Automatic Control Description 自动控制说明
The HVAC system is controlled locally by PLC. 空调系统设置就地PLC控制。
Mode determination: Based on the temperature and humidity sensors on the fresh air duct, calculate the air enthalpy or moisture, and automatically determine the conversion between summer, winter, and transitional season working modes.
模式判断：根据新风管道上温度传感器，计算出新风焓值或含湿量，自动判断转换夏季/冬季/过渡季节工作模式。
Temperature and humidity control: Summer condition: 温、湿度控制; 夏季工况; Control the opening degree of the control valve on the chilled water pipe according to the room temperature and humidity, and control the room temperature and humidity within the design value.
When there is a contradiction between humidity and temperature control, prioritize humidity control. If controlling humidity causes the outlet temperature to be too low, adjust the control valve on the hot water pipe to control the outlet temperature within the design value.
根据房间内温、湿度传感器，控制表冷器上的电动阀的开度，控制房间温、湿度在设计值。
当湿度和温度控制发生矛盾时，以控制湿度优先，如果因为控制湿度而使出水温度过低时，调节加热热电动阀以控制出水温度在设计值。
Winter condition: 冬季工况; Based on the room temperature, the opening degree of the electric valve in the reheat section of the air conditioning unit is controlled to keep the room temperature at the designed value.
根据房间内温度传感器，控制空调机组再热器电动阀的开度，控制房间温度在设计值。
Based on the room humidity, the opening degree of the electric valve in the preheating section is controlled to maintain the relative humidity in the room at the designed value. 根据房间内湿度传感器，预热器电动阀的开度，控制房间内相对湿度在设计值。
Transition season operation condition: 过渡季工况; Based on the room temperature and humidity, the electric valves of the chiller and hot water pipe are controlled to maintain the indoor temperature and humidity at the designed values.
根据房间内温、湿度传感器，控制表冷器及加热器电动阀，控制室内温度湿度在设计值。
Positive pressure control: 正压控制; The opening degree of the variable air volume damper in the exhaust duct is controlled by the indoor pressure differential sensor, and the exhaust volume is adjusted to meet the indoor pressure requirements.
根据室内压差传感器控制排风风管变风量阀开度，调节排风量，以满足室内压力要求。
Air volume control: 风量控制; The variable-frequency exhaust fan operates based on the signals from the indoor pressure sensor and the variable air volume valve for exhaust. When there is a conflict between the air volume control and the pressure control, the pressure control takes precedence. The MAU main supply duct maintains a constant static pressure and delivers a constant air volume for ventilation.
变频排风机根据室内压差传感器信号及排风变风量阀信号变频运行，当风量控制与压力控制发生矛盾时，以压力控制优先。MAU主送风管定静压恒定送风量送风。
Filter differential pressure alarm: the primary differential pressure > 100 Pa, the intermediate differential pressure > 250 Pa, the high-efficiency differential pressure > 350 Pa, an alarm will be triggered.
过滤器压差报警：当初效压差大于100Pa，报警，当中效压差大于250Pa，报警，当高效压差大于350Pa，报警，报警值可调。
Startup sequence: Fresh air electric damper → Supply fan → System exhaust fan. Shutdown sequence is the reverse.
开机顺序：新风电动阀→送风机→系统排风机；关机顺序相反。
Interlocking and Detection: 联锁及检测; The fire damper of the air supply main pipe of the air conditioner is interlocked with the air conditioner. When the fire damper is closed, the air conditioner is also turned off. 空调器送风总管防火阀与空调器联锁，防火阀关，空调器关。
The electrical on-off fresh air damper of the air conditioner is interlocked with the fan of the air conditioner. When the fresh air damper is open, the fan is turned on; when the fresh air damper is closed, the fan is turned off; and when the fresh air damper is closed, the air conditioner is also turned off. 空调器新风电动风阀与空调器风机联锁，新风阀开，风机开，风机关，新风阀关。
The automatic conversion of operating conditions is carried out to determine the air treatment mode (dehumidification, cooling, heating, humidification, etc.). 自动转换工况点，以便确定空气处理方式（去湿、降温、加热、加湿等）。
The supply/exhaust fan is set with pressure difference control: when the pressure difference is 0, an alarm is triggered.
送、排风机设置压差控制：当压差为0时，报警。
The parameters that require inspection and recording include: outdoor temperature, humidity, indoor temperature, humidity.
需要记录记录的参数有：室外温度、湿度、室内温度、湿度。
All the position signals of the electric valves are fed back to the upper computer. 所有电动风阀位置信号反馈至上位机。

(A1)



Abbreviation 缩写	Legend 图例	Description 说明
AD		Square Diffuser 方形散流器
SG		Single Deflection Grille 单层百叶风口
L		Rain proof Louver 防雨百叶风口
WL		Exterior Walls Rainproof Louver 外墙防雨百叶
		Multi-blade Damper 手动对开多叶调节阀
		Check Damper 止回阀
MD		Motorized On-off Damper 电动开关密闭阀
FD (FVD)		Normally Open 70°C Fire Damper 70°C常开防火(调节)阀



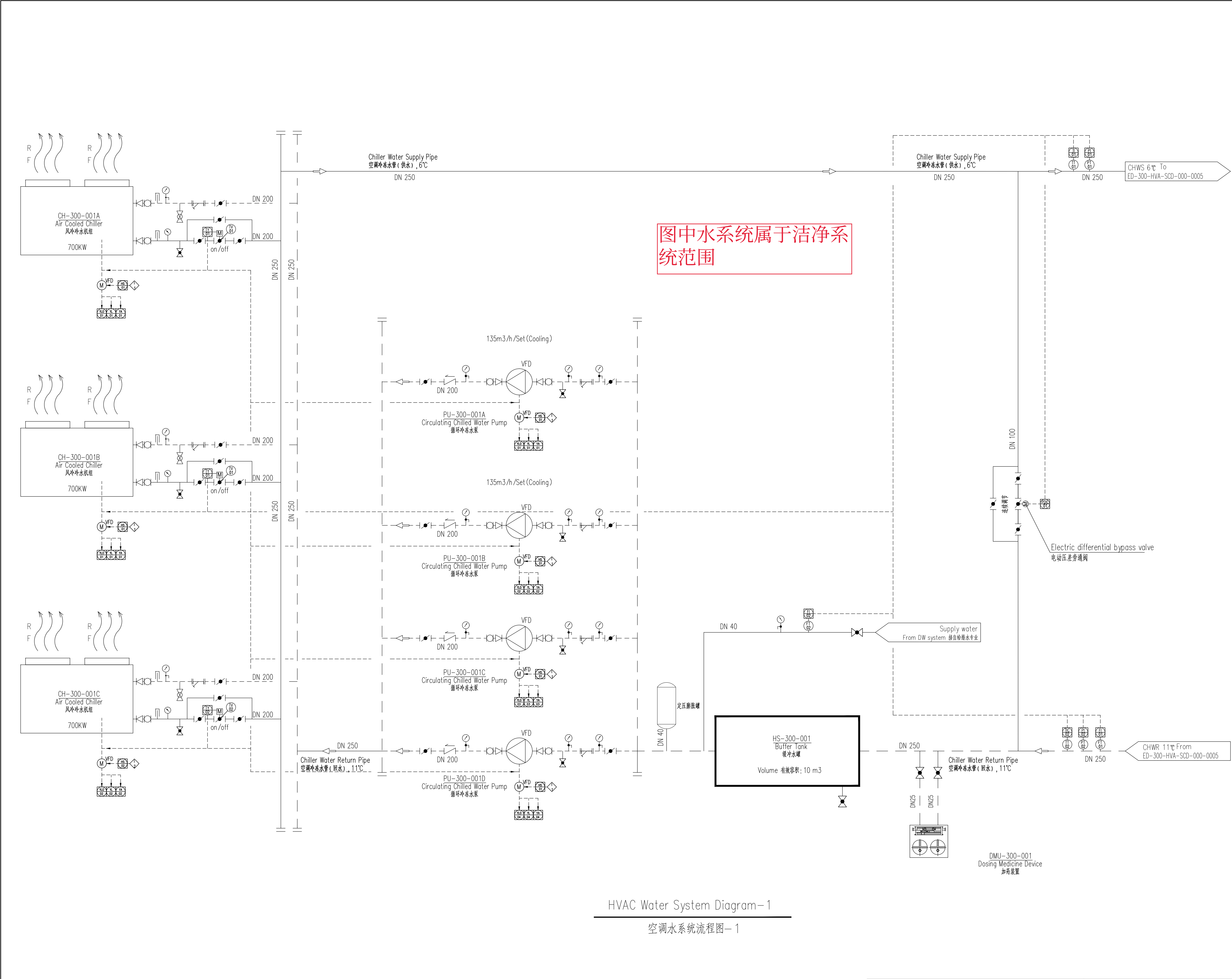
NOTES 备注

- Wall axial flow fan is set for the future area.
预留区域设置边墙风机通风。
- The air lock 6 is equipped with a positive pressure ventilation system. It uses side wall supply fans for positive pressure ventilation, with one fan in operation and another as a backup. Positive pressure indication instruments and pressure loss alarm devices are installed indoors, and they are interlocked with the positive pressure ventilation system. When the relative positive pressure value between the indoor area and the adjacent room is lower than 25Pa and remains so for 1 minute, an alarm signal should be issued, and the standby fan SF-300-201B should be automatically put into operation. Ensure that the indoor pressure is higher than that of the surrounding rooms by more than 30Pa.
门斗六设置正压送风系统，采用边墙送风机正压送风，风机一开一备。室内设置正压指示仪表及失压报警装置，并与正压送风系统联锁。当室内与临室内相对正压值低于25Pa持续1min后，应发出报警信号，并使备用送风机自动打开，保证室内压力高于周边房间30Pa以上。

HVAC System Flow Diagram for Normal Area

普通区域空调系统流程图

						DATE		MADDE		TITLE
						SCALE	N/A	CH•XD		HVAC System Flow Diagram for Normal Area
						JOB No.		APP•D		
SECURITY LEVEL LEVEL-2	No.	DESCRIPTION	DATE	MADDE	CH•XD	APP•D	DWG. No. ED-300-HVA-SCD-000-0003			
REVISIONS							REV.			



HVAC Water System Diagram-1

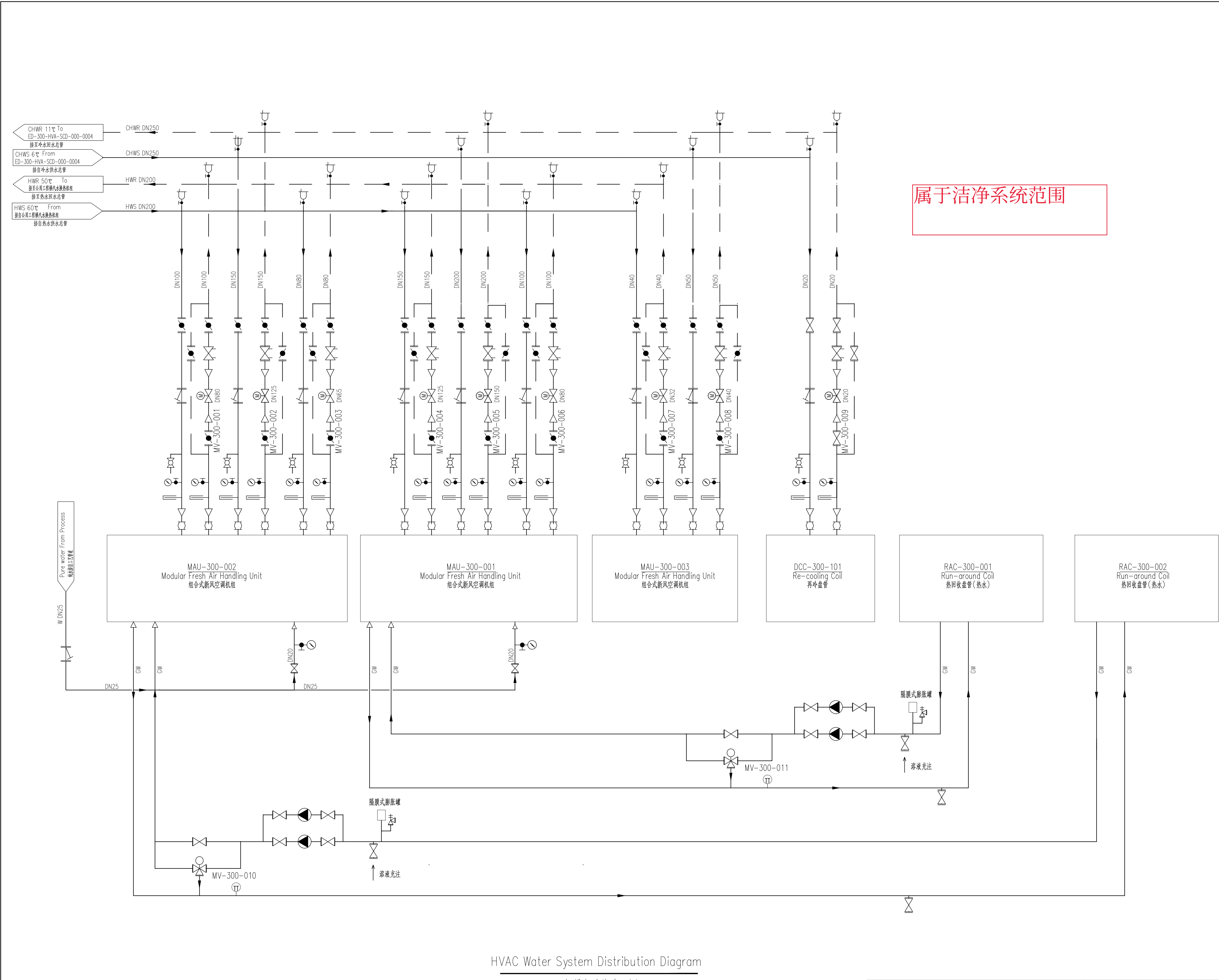
空调水系统流程图-1

LEGEND 图例			
Abbreviation 缩写	Legend 图例	Description 说明	
MV		Electric two-way valve	电动二通阀
		Globe valve	截止阀
		Ball valve	球阀
		Butterfly valve	蝶阀
		Check valve	止回阀
		Pressure gauge	压力表
		Thermometer	温度计
		Y type filter	Y型过滤器
		Plug valve	旋塞阀
		Flow meter	流量计
		Water pump	水泵
		Rubber hose	软接管
		Reducer	变径
		Temperature transmitter	温度传感器
		Pressure transmitter	压力传感器
		Flow meter	流量计
		Local field mounted instrument	就地现场仪表
	-----	Control line	控制线

NOTES 说明

- Each chiller unit is equipped with its own control cabinet, which has the functions of controlling the unit itself and interlocking with the chilled water pump and electric butterfly valve, as well as monitoring the operating parameters.
每台冷水机组均自带控制柜，具有对本机的控制及对匹配的冷水泵、电动蝶阀联锁的功能，和对运行参数的监测功能。
- Startup: Electric butterfly valve - Chilled water pump - Chiller.
Shutdown sequence is the reverse of the startup sequence.
冷水机组启动顺序：电动蝶阀→冷水泵→冷水机组。停机顺序与启动顺序相反。
- All the chillers are centrally controlled by a group control system. They can automatically start and stop according to changes in cooling and heating loads, and perform unit control and load distribution to optimize system operation. The chilled water pumps are variable-frequency pumps. The frequency of the chilled water pumps is controlled based on the pressure difference between the supply and return water in the most unfavorable loop. When the pressure difference is less than the set value of the pump, the pump speed is increased; otherwise, the pump speed is decreased. When the frequency adjustment reaches the minimum flow rate allowed by a single chiller, the chiller side and the chilled water pumps will no longer be adjusted. Instead, the electric bypass valves on the supply and return water main pipes will be used for adjustment to ensure that the minimum flow rate of the chiller is the sum of the water volume on the load side and the flow volume of the bypass pipe.
全部冷水机组集中采用群控系统方式，可根据冷热负荷的变化，自动开停机，进行台数控制和负荷分配，优化系统运行；冷冻水泵为变频泵，根据最不利环路的供回水压差控制冷冻水泵变频。压差小于水泵设定值时，则提高水泵的转速，反之降低水泵的转速。当变频调节至单台冷水机组允许的最小流量时，冷水机组侧及水泵不再调节，则通过供回水主管上的电动旁通阀调节，使冷水机组的最小流量为负荷侧水量与旁通管流量之和。
- The operating status, main working parameters and alarm signals of the chilled water system equipment are all connected to the host computer for unified monitoring. The monitoring points include, but are not limited to:
 - Monitoring of all equipment failures, start-stop status.
 - Monitoring of the temperature, pressure and flow of the supply and return water pipes of the chilled water and hot water circulation pumps.
 - Setting pressure differential bypass for the supply and return water pipes of chilled water and hot water, with the set value adjustable on-site.
 - Interlock control of the electric butterfly valves of the chilled water and hot water return pipes of the chiller with the chiller.
 - Centralized monitoring of the opening and closing status of the electric valves. If there are any faults such as inability to open or close, there should be an alarm display.冷水水设备的运行状态及主要工作参数及报警信号均引至上位机集中监控，包括但不限于以下监控点：
 - 所有设备的故障，启停状态监控。
 - 冷冻水循环泵供回水主管的温度、压力、流量监测。
 - 冷冻水供回水主管设置压差旁通，设定值现场可调。
 - 冷机冷冻水回水管电动蝶阀与冷水机联锁控制。
 - 集中监测电动阀的开启和关闭状态，如果出现无法打开或者关闭故障，应该有报警显示。

SECURITY LEVEL LEVEL-2	△						DATE		MADE		TITLE HVAC Water System Diagram
	△a						SCALE	N/A	CH*KD		
	△b	Issue for Inquiry	May/04/26	X1R	X4H	NWQ	JOB No.		APP*D		
	△c	BED PACKAGE	Apr/30/26	X1R	X4H	NWQ					
	No.	DESCRIPTION	DATE	MADE	CH*KD	APP*D	DWG. No. ED-300-HVA-SCD-000-0004				REV. △a
	REVISIONS										



属于洁净系统范围

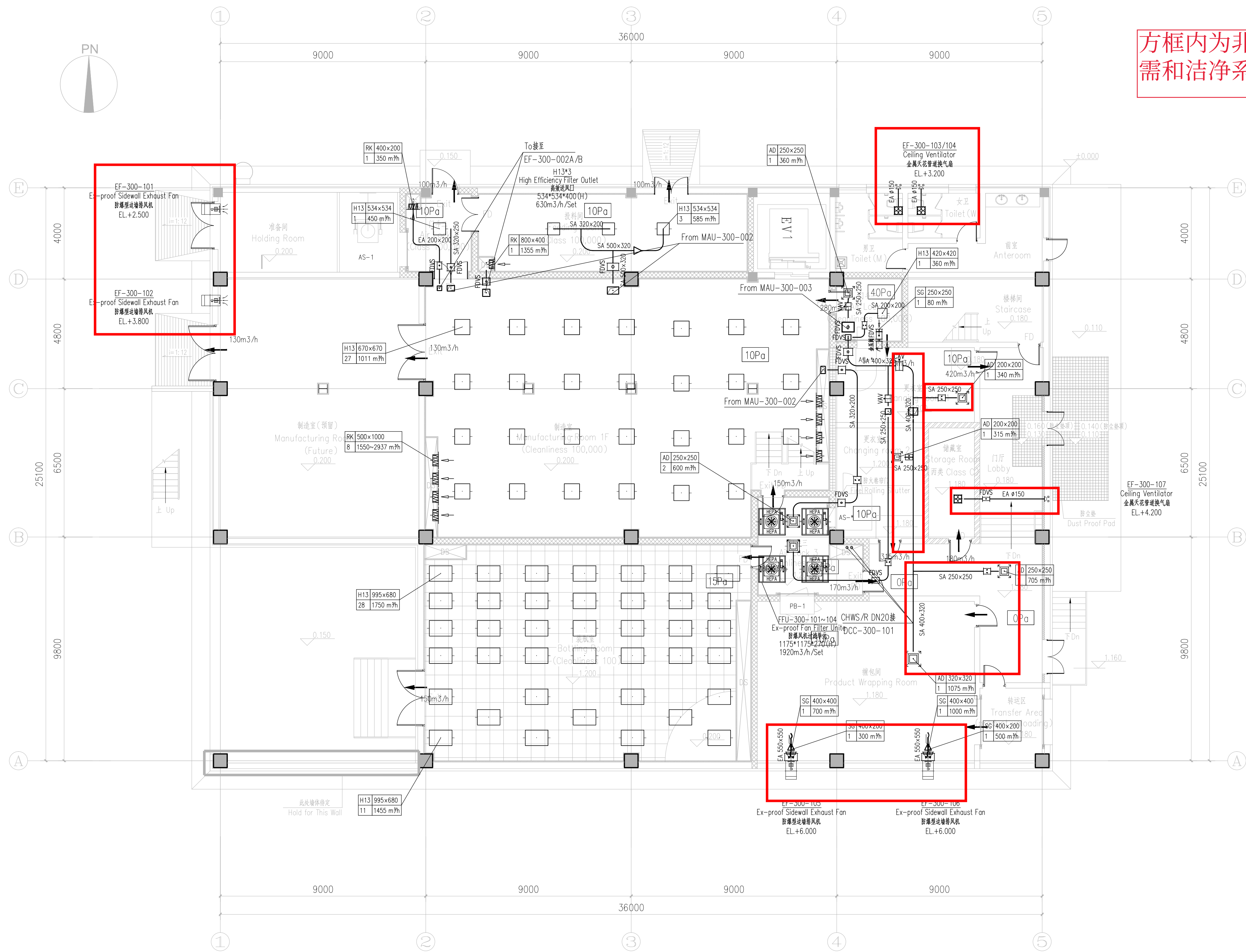
LEGEND 图例			
Abbreviation 缩写	Legend 图例	Description 说明	
		Electric two-way valve	电动二通阀
		Electric three-way valve	电动三通阀
		Globe valve	截止阀
		Ball valve	球阀
		Butterfly valve	蝶阀
		Balance valve	平衡阀
		Pressure gauge	压力表
		Thermometer	温度计
		Y type filter	Y型过滤器
		Automatic exhaust valve	自动排气阀
		Plug valve	旋塞阀
		Reducing valve	减压阀
CHWS		Chilled water supply pipe	冷水供水管
CHWR		Chilled water return pipe	冷水回水管
HWS		Hot water supply pipe	热水供水管
HWR		Hot water return pipe	热水回水管
GW		Glycol water pipe	乙二醇溶液 供/回水管 (热回收)
AC		Condensate water pipe	冷凝水管
Z		Steam pipe	蒸汽管
C		Steam condensate pipe	蒸汽凝水管
W		Humidifier pipe	加湿水管

NOTES 说明

HVAC Water System Distribution Diagram

空调水系统分配图

SECURITY LEVEL LEVEL-2	REVISIONS					DATE	N/A	MADE		TITLE	
	No.	DESCRIPTION	DATE	MADE	CH·KD	APP·D	JOB No.	APP·D		HVAC Water System Distribution Diagram	
										DWG. No.	
										ED-300-HVA-SCD-000-0005	



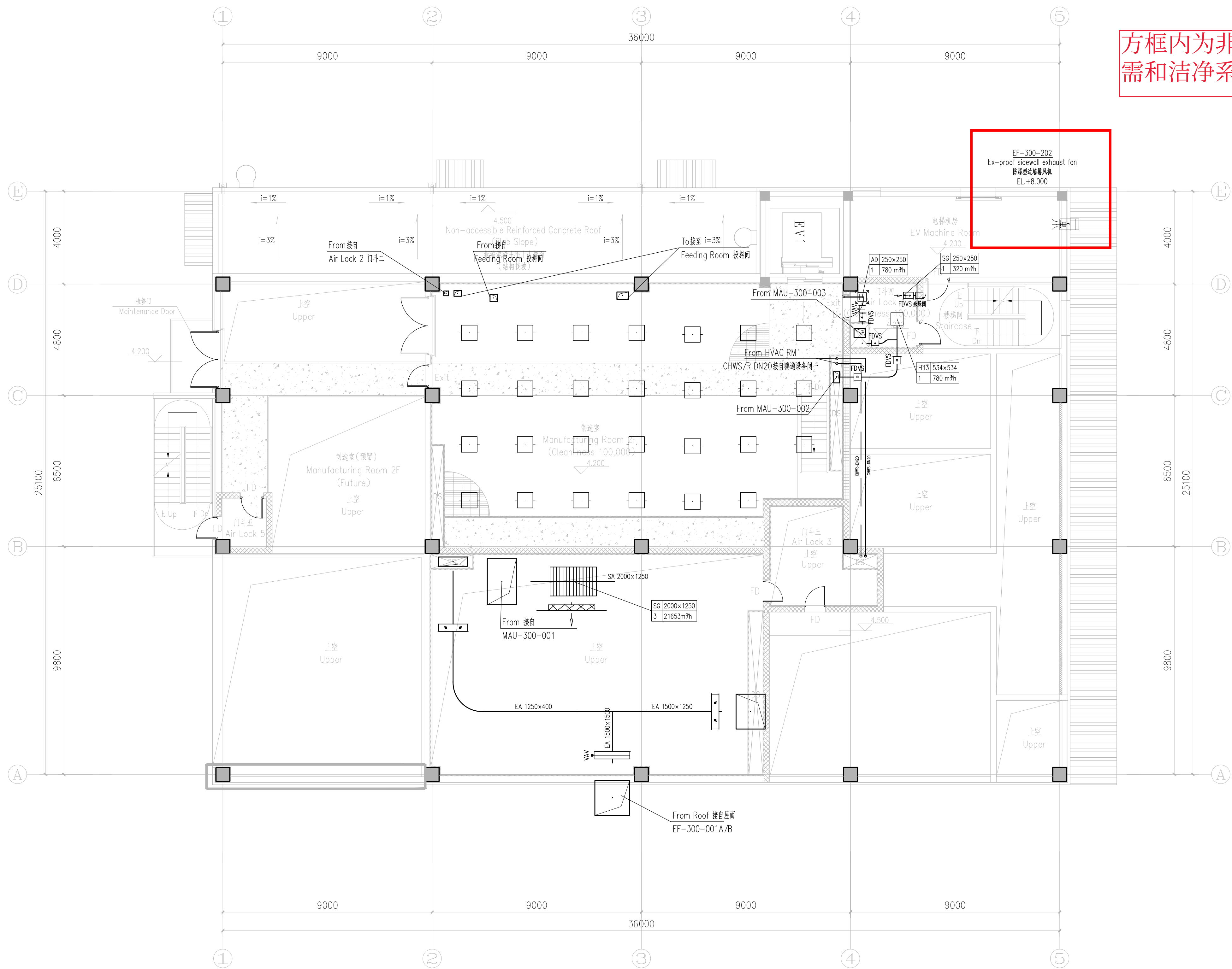
方框内为非洁净区系统
需和洁净系统分开报价

NOTES 备注

1. The width of the door gaps for entering and exiting the clean room through the clean doors and positive pressure airlock doors is 2mm. The door gap width for the emergency evacuation doors in the clean area is calculated as 1mm, while the door gap width for non-clean area doors is 3mm.
洁净室日常进出洁净门及正压门日常进出门口门缝宽度2mm, 洁净室逃生疏散门门缝宽度按1mm计, 非洁净区门缝按3mm计。
The Future area is not in the scope of work.
Future区域不在本次设计范围内。

HVAC Layout for EL+0.200
EL+0.200 暖通系统布置图

SECURITY LEVEL LEVEL-2	No.	DESCRIPTION	DATE	MADE	CH·XD	APP·D	△				DATE		MADE		TITLE			
							△				SCALE	1/100	CH·XD		Manufacturing Building			
							△				JOB No.		APP·D		HVAC Layout for EL+0.200			
							△											
REVISIONS										DWG. No.		REV.						
										ED-300-HVA-LAY-000-0001		△						

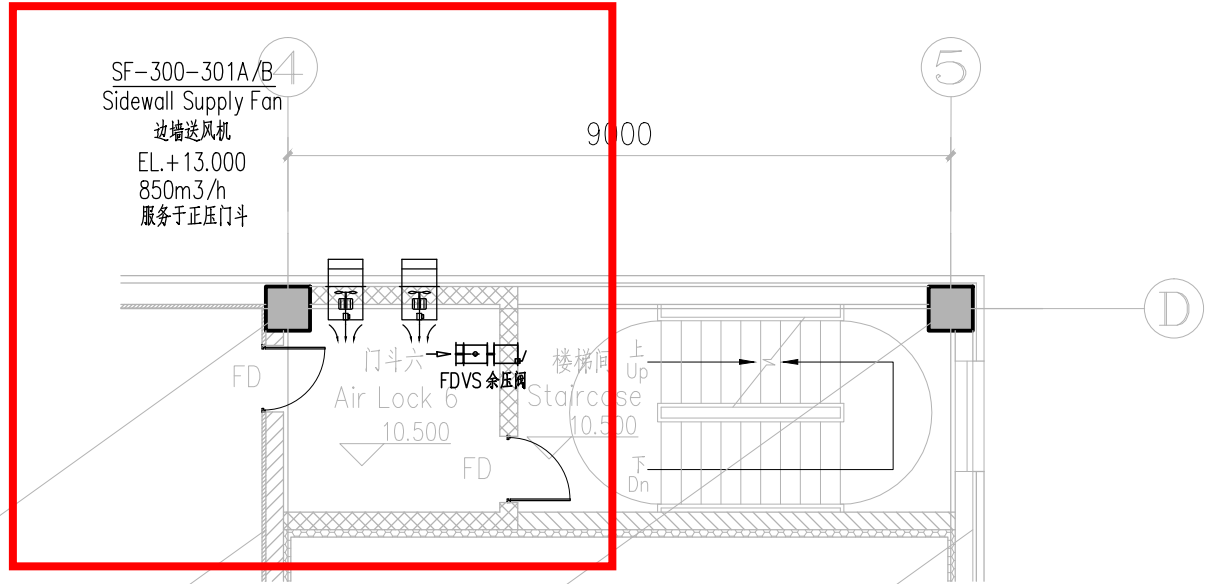
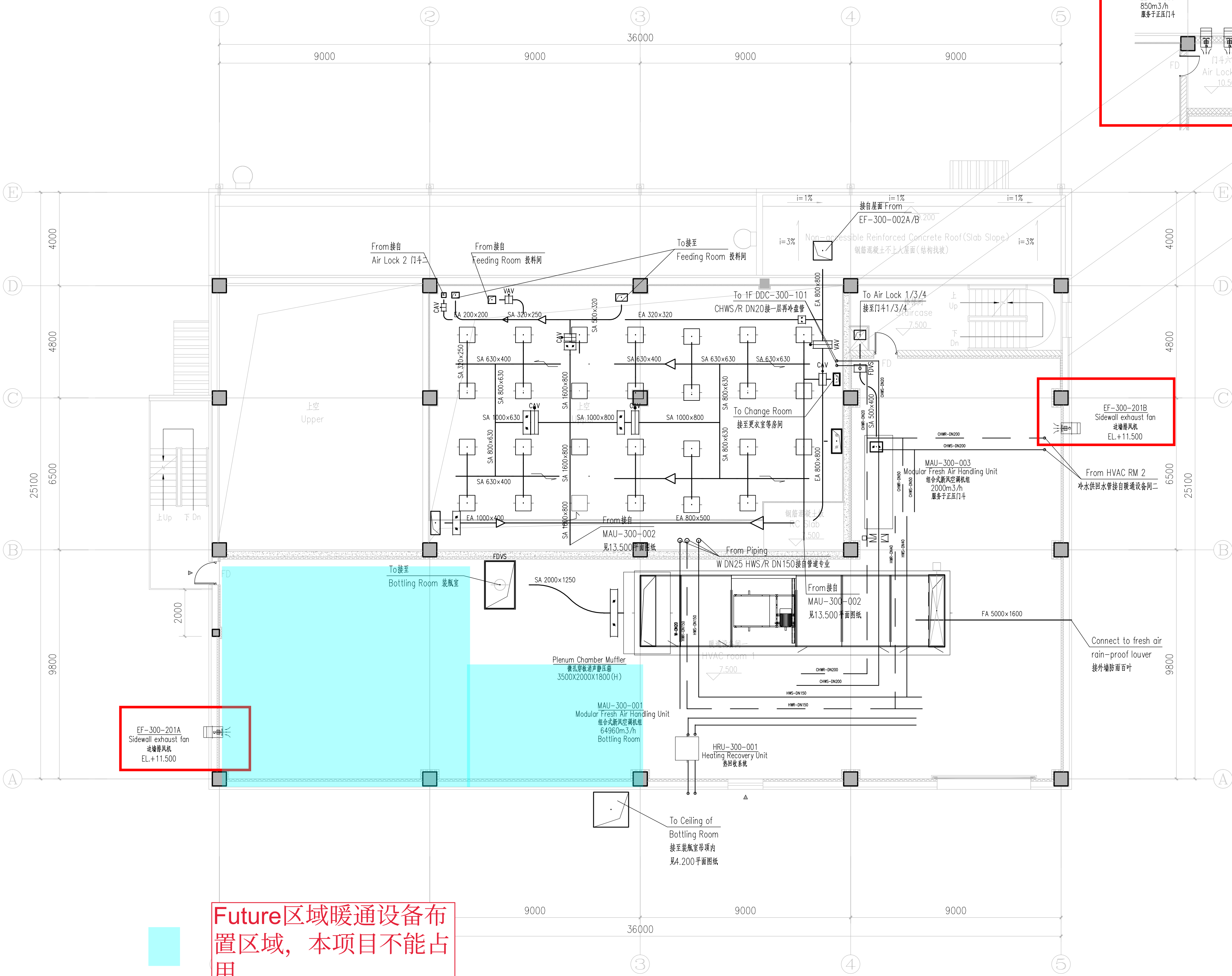


方框内为非洁净区系统
需和洁净系统分开报价

- NOTES 备注
- The width of the door gaps for entering and exiting the clean room through the clean doors and positive pressure airlock doors is 2mm. The door gap width for the emergency evacuation doors in the clean area is calculated as 1mm, while the door gap width for non-clean area doors is 3mm.
洁净室日常进出洁净门及正压门日常进出门门缝宽度2mm, 洁净区逃生疏散门门缝宽度按1mm计, 非洁净区门缝按3mm计。
 - The Future area is not in the scope of work.
Future区域不在本次设计范围内。

HVAC Layout for EL+4.200
EL+4.200 暖通系统布置图

SECURITY LEVEL LEVEL-2							DATE	1/100	MADE		TITLE Manufacturing Building HVAC Layout for EL+4.200	REV. 
							SCALE		CH*XD			
							JOB No.		APP'D			
		RED PACKAGE Apr/30/26 X1R X4H NWO										
	No.	DESCRIPTION	DATE	MADE	CH*XD	APP'D	DWG. No. ED-300-HVA-LAY-000-0002					
REVISIONS												



方框内为非洁净区系统
需和洁净系统分开报价

EF-300-201A
Sidewall exhaust fan
进墙排风机
EL+11.500

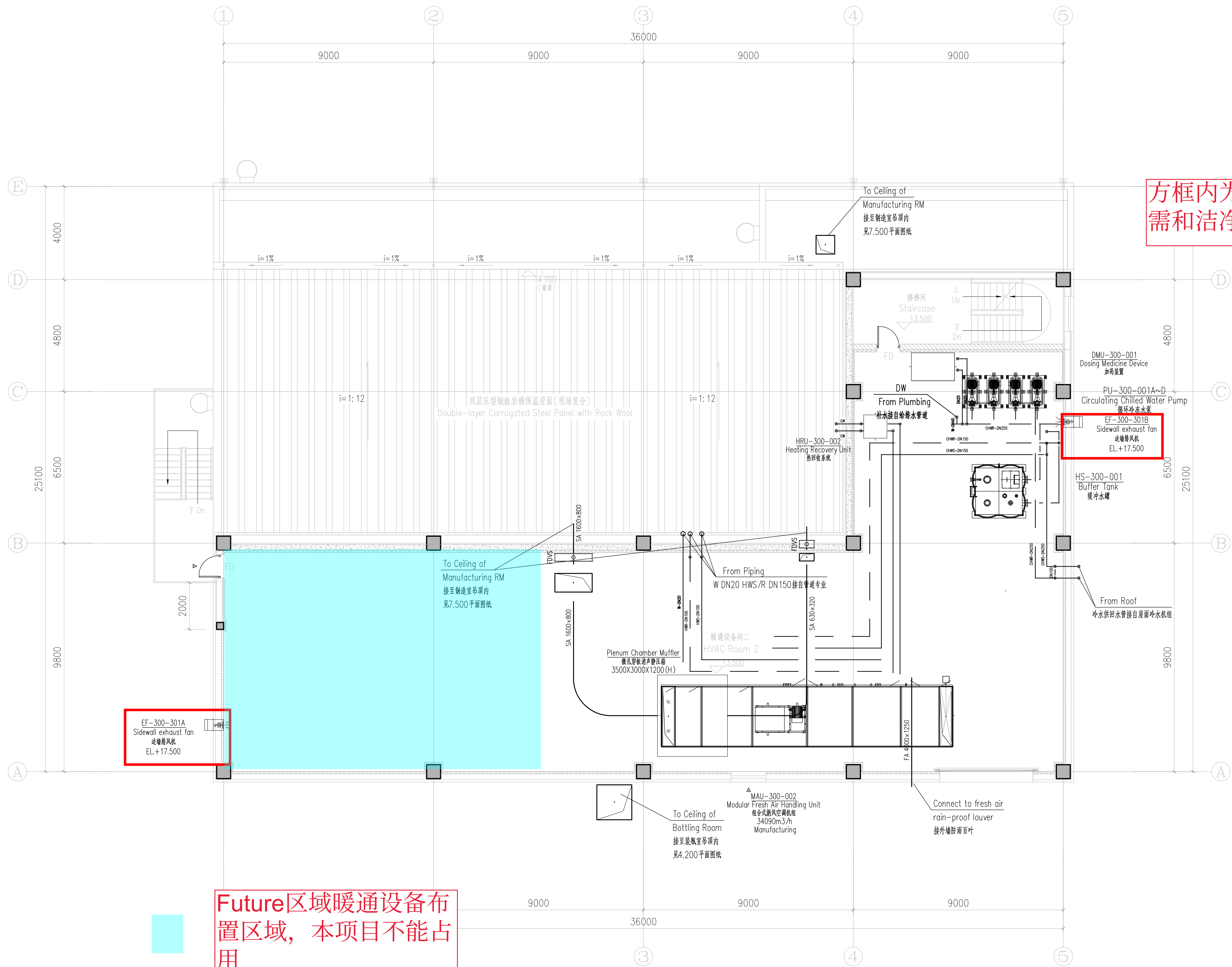
EF-300-201B
Sidewall exhaust fan
进墙排风机
EL+11.500

Future区域暖通设备布
置区域，本项目不能占用

- NOTES 备注
- At this stage, the water pipe system is only used to indicate the main pipe locations. For the piping connections of the air conditioning equipment, please refer to the air and water system diagrams. The piping layout will be further detailed in the detailed design stage.
本阶段水管系统仅示意主管位置，空调设备管口配管可详见风、水系统图，具体水管敷设在详设阶段深化。
 - The Future area is not in the scope of work.
Future区域不在本次设计范围内。

HVAC Layout for EL+7.500
EL+7.500 暖通系统布置图

SECURITY LEVEL LEVEL-2	REVISIONS					DATE	1/100	MADE		TITLE Manufacturing Building HVAC Layout for EL+7.500 DWG. No. ED-300-HVA-LAY-000-0003
	No.	DESCRIPTION	DATE	MADE	CH·XD	APP·D				



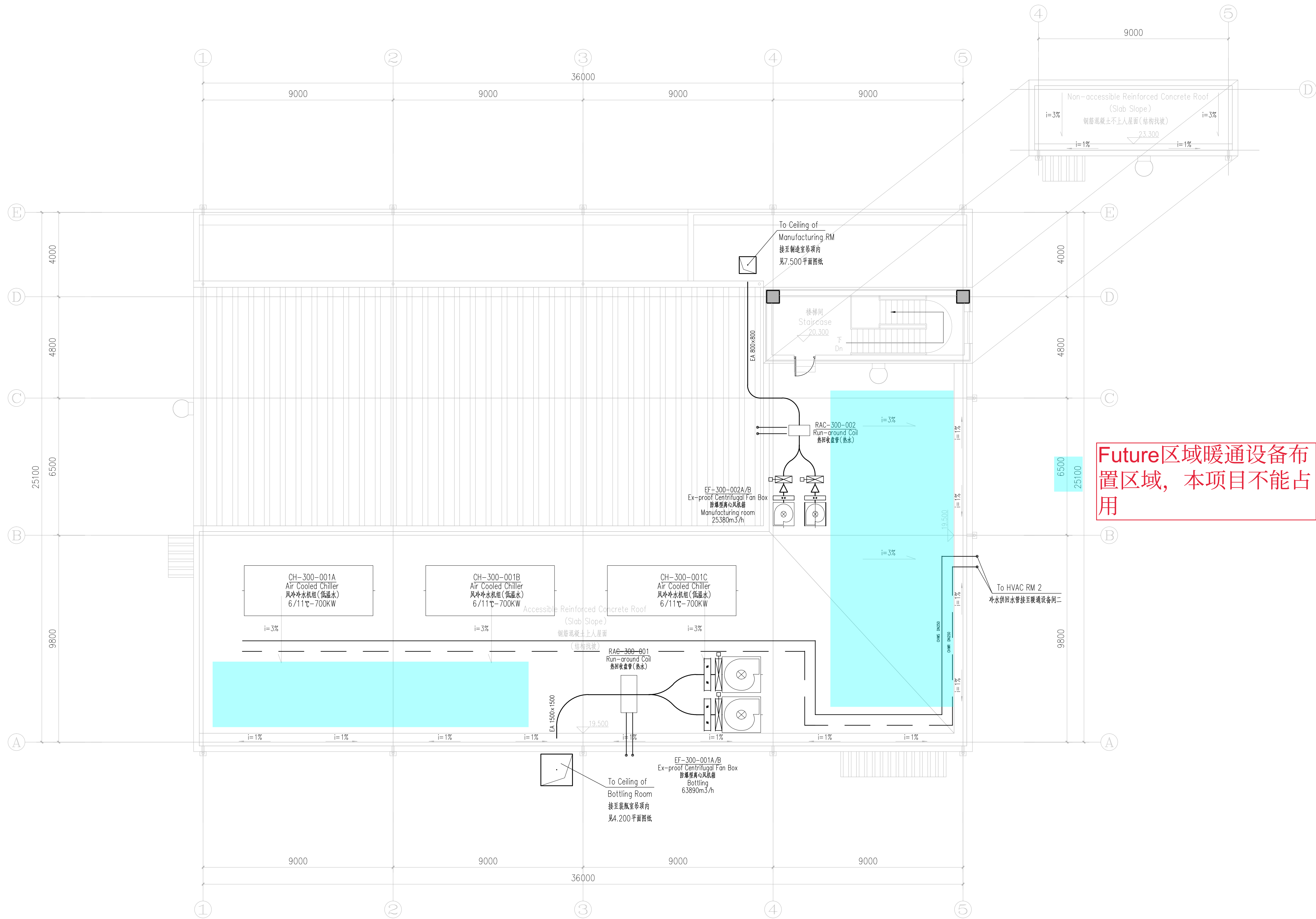
Future区域暖通设备布置区域，本项目不能占用

方框内为非洁净区系统需和洁净系统分开报价

- NOTES 备注
- At this stage, the water pipe system is only used to indicate the main pipe locations. For the piping connections of the air conditioning equipment, please refer to the air and water system diagrams. The piping layout will be further detailed in the detailed design stage. 本阶段水管系统仅示意主管位置，空调设备接口配管可详见风、水系统图，具体水管敷设详见设计阶段深化。
 - The Future area is not in the scope of work. Future区域不在本次设计范围内。

HVAC Layout for EL+13.500
EL+13.500 暖通系统布置图

SECURITY LEVEL LEVEL-2	△ △ △ △					DATE		MADE		TITLE Manufacturing Building HVAC Layout for EL+13.500		REV. △
						SCALE		1/100				
						JOB No.		APP•D				
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BED PACKAGEApr/30/08XRXHXHWNQ												
DWG. No. ED-300-HVA-LAY-000-0004												



HVAC Layout for Roof
屋面暖通系统布置图

- NOTES 备注
- At this stage, the water pipe system is only used to indicate the main pipe locations. For the piping connections of the air conditioning equipment, please refer to the air and water system diagrams. The piping layout will be further detailed in the detailed design stage.
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SECURITY LEVEL LEVEL-2	No.	DESCRIPTION	DATE	MADE	CH-KD	APP-D	DATE		MADE		TITLE				
							SCALE		1/100		CH-KD		Manufacturing Building HVAC Layout for Roof		
							JOB No.		APP'D						
							REVISIONS		DWG. No. ED-300-HVA-LAY-000-0005						
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